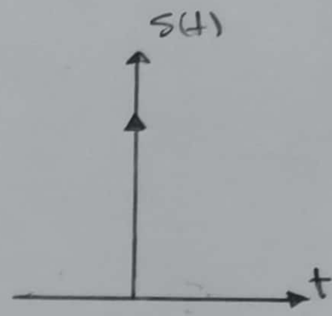
"Some Important signals"

one: Unit Impulse signal :-



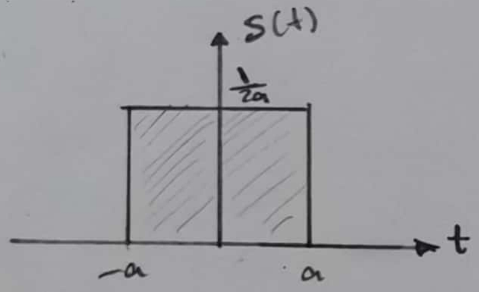
→ Dirac Delta function.

→ $\delta(t)$ where $\delta(t) = \begin{cases} \infty , & t=0 \\ 0 , & t \neq 0 \end{cases}$

→ Area of unit impulse function is always equal to "1".

$$\int_{-\infty}^{\infty} \delta(t)\, dt = 1$$

→ $\delta(t)$ Can be represent as $\begin{cases} 1/2a & -a<t<a \\ 0 & \text{otherwise} \end{cases}$



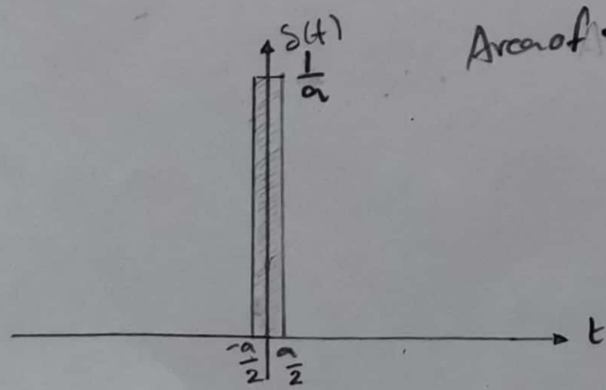
$$\text{Area } \delta(t) = \int_{-\infty}^{\infty} \delta(t)\, dt$$

$$= \int_{-a}^{a} \frac{1}{2a}\, dt$$

$$= \frac{1}{2a}\left[t\right]_{-a}^{a} \Rightarrow \frac{1}{2a}\left[a+a\right]$$

$$= \frac{1}{\cancel{2a}} * \cancel{2a} \Rightarrow \text{Area } \delta(t) = 1$$

→ IN fact $a \to 0$ so $\frac{1}{a} \to \infty$

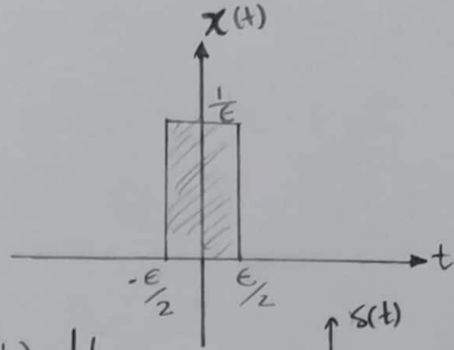


$$\text{Area of } \delta(t) = \text{Length} \times \text{width}$$

$$= \frac{1}{a} \times a$$

$$= 1$$

"Properties of Impulse Signal"

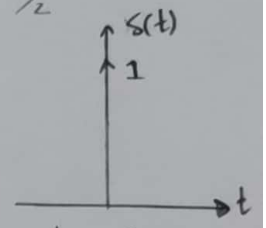


① $\int_{-\infty}^{\infty} \delta(t) \cdot dt = 1$

where $\delta(t) = \lim_{\epsilon \to 0} x(t) = \begin{cases} \infty & t = 0 \\ 0 & t \neq 0 \end{cases}$

Area of $\delta(t) = \int_{-\infty}^{\infty} \delta(t) \cdot dt = \int_{-\infty}^{\infty} \lim_{\epsilon \to 0} x(t) \cdot dt$

$= \lim_{\epsilon \to 0} \left[\overset{1}{\int_{-\infty}^{\infty} x(t) \cdot dt} \right] \Rightarrow \lim_{\epsilon \to 0} 1 = 1$

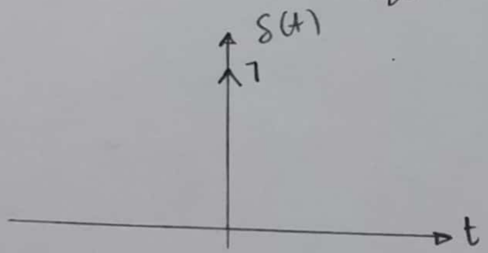


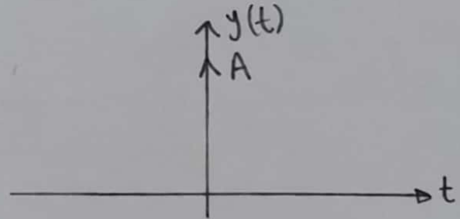
when $\epsilon \to 0$

so $\boxed{\text{Area of } \delta(t) = 1}$

② weight or strength of an impulse: Let $y(t) = A\,\delta(t)$

to Compute Area of $y(t) = \int_{-\infty}^{\infty} y(t) \cdot dt = A \int_{-\infty}^{\infty} \delta(t) \cdot dt = A$ ⇛ weight or Area of impulse signal.

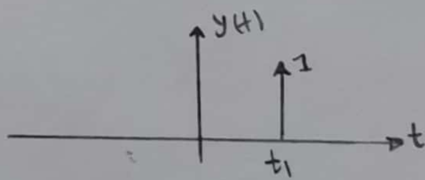




③ Unit impulse is Even signal because $\delta(t) = \delta(-t)$

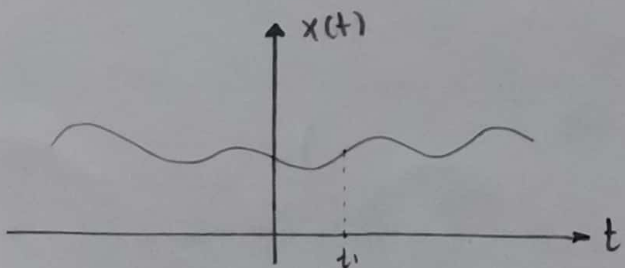
④ " " is NENP " " when $t = 0 \Rightarrow \delta(t) = \infty$

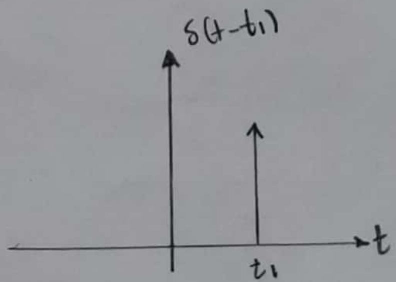
⑤ shifting of $\delta(t)$: Let $y(t) = \delta(t - t_1)$

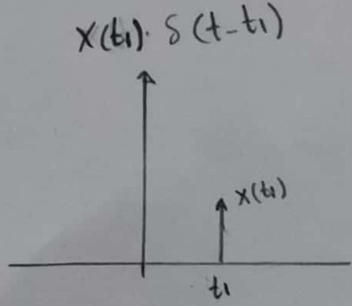


⑥ Multiplication:

$$x(t) \cdot \delta(t - t_1) = x(t_1)\,\delta(t - t_1)$$





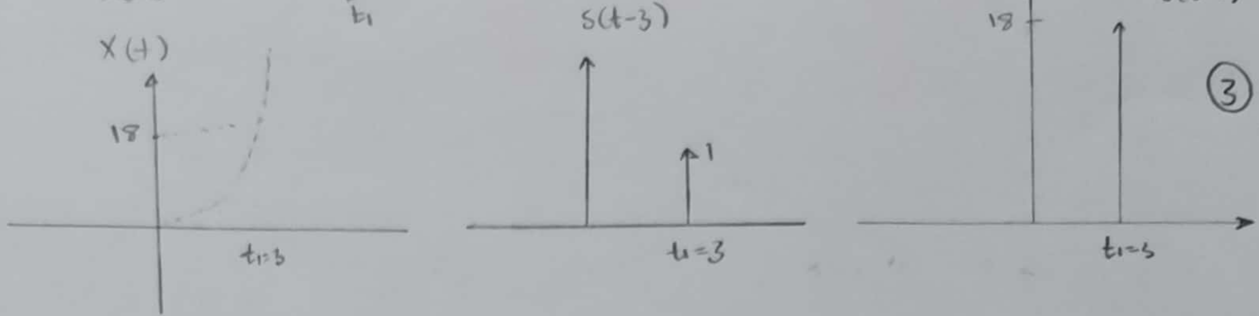


EX: $y(t) = \underset{\sim}{2t^2} \cdot \delta(t - \underset{t_1}{3})$ plot $y(t)$?

$y(t) = x(t_1)\,\delta(t - t_1)$

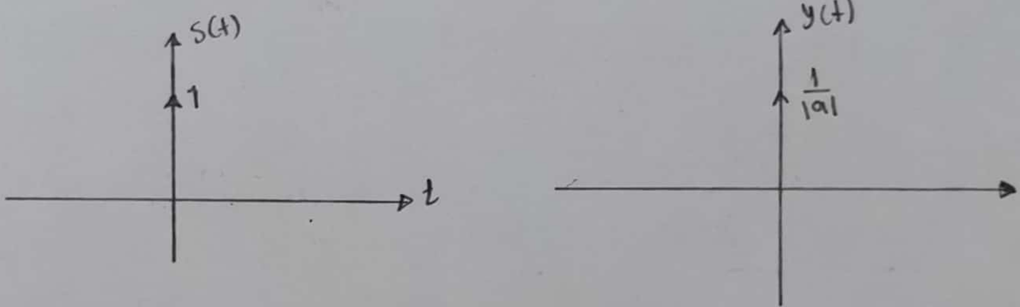
$18\,\delta(t-3)$

③



⑦ time scaling $y(t) = \delta(at) = \frac{1}{|a|}\delta(t)$ where $a \neq 0$

note: Scaling time by (a) make the Area $= \frac{1}{|a|}$ because (a) act as Compression the x-axis by (a) and that make the total Area of $y(t) = 1 \times \frac{1}{a}$ so $y(t) = \frac{1}{a}\delta(t)$



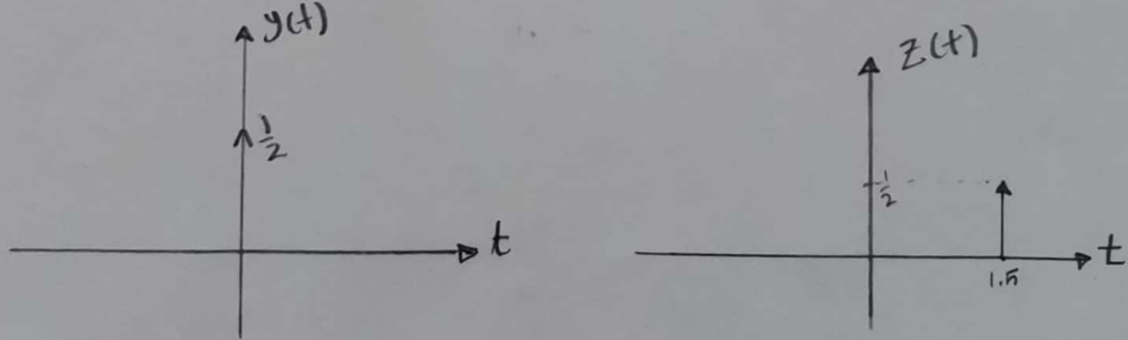
E.X:
① $y(t) = \delta(-2t)$ ② $Z(t) = \delta(-2t+3)$

plot $y(t)$ and $Z(t)$

$y(t) = \delta(-2t) = \frac{1}{|-2|}\delta(t) = \frac{1}{2}\delta(t)$

$Z(t) = \delta(-2t+3) = \delta(-2(t-1.5))$

$= \frac{1}{|-2|}\delta(t-1.5) \Rightarrow Z(t) = \frac{1}{2}\delta(t-1.5)$



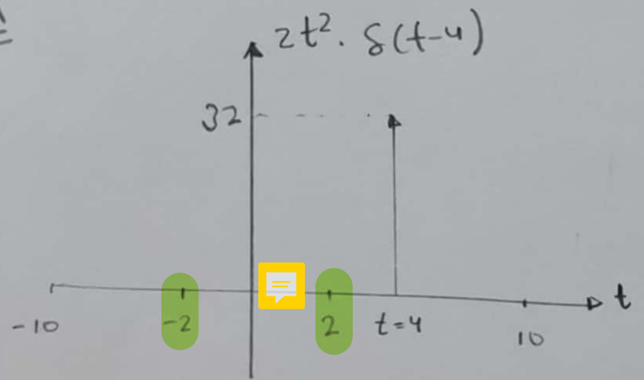
H.W Plot $y(t) = \cos(4t)\cdot\delta(2t-\pi)$ Ans: $\frac{1}{2}\delta(t-\pi/2)$

⑧ $\int_{-\infty}^{\infty} x(t)\cdot\delta(t-t_1)\,dt \Rightarrow \int_{-\infty}^{\infty} x(t_1)\,\delta(t-t_1)\,dt$ ④

$= x(t_1)\underbrace{\int_{-\infty}^{\infty}\delta(t-t_1)\,dt}_{1} \Rightarrow \int_{-\infty}^{\infty} x(t)\cdot\delta(t-t_1)\,dt = x(t_1) =$ Constant.

E.x Compute ① $\int_{-10}^{10} 2t^2\cdot\delta(t-4)\,dt$ ② $\int_{-2}^{2} 2t^2\cdot\delta(t-4)\,dt$

Sol



$t=4 \Rightarrow 2t^2 \Rightarrow 2(4)^2$
$= 2\times16$
$= 32$

$\Rightarrow \int_{-10}^{10} 32\,\delta(t-4)\,dt = 32$, but $\int_{-2}^{2} 32\,\delta(t-4)\,dt =$ Zero

H.W Compute the following Integration:

① $\int_{-5}^{4} \delta(t-5)\,dt$ Ans = Zero

② $\int_{-5}^{4} \delta(t-2)\,dt$ Ans = 1

③ $\int_{-\infty}^{\infty} \sin(t)\cdot\delta(-2t+1)\,dt$ Ans $= \frac{1}{2}$

④ $\int_{-\infty}^{\infty} e^{-2t}\cdot\delta(-2t+1)\,dt$ Ans $= \frac{1}{2}e^{-1}$

⑤ $\int_{-\infty}^{\infty}\left[\cos(\pi t)\cdot\delta(t-2) + 3\,\delta(t+1) + \sin(\pi t)\cdot\delta(2t-1)\right]dt$

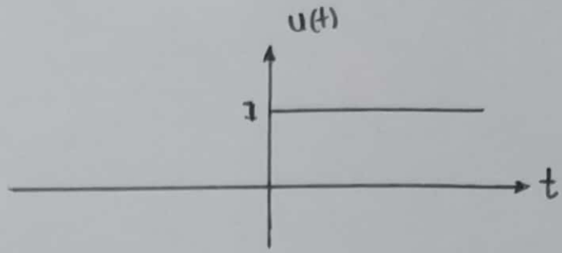
Ans $= 1 + 3 + \frac{1}{2} = 4.5$

⑥ $\int_{-\infty}^{\infty} e^{-t}\,\delta(2t-2)\,dt$

Two: unit step signal :-

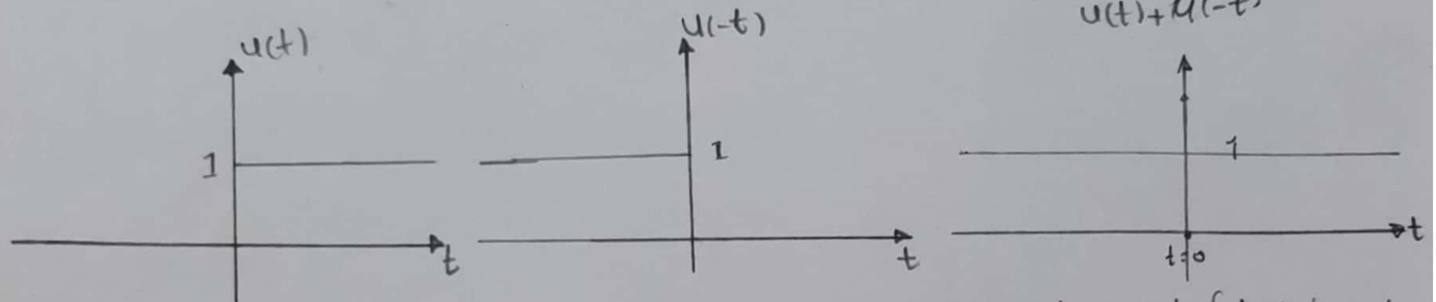
→ Heavi side step function another name of unit step signal.

→ $u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$



Properties of unit step signal :-

① $u(t) + u(-t) = 1$



t=0 Point of discontinuity

at $t=0 \Rightarrow u(t) = 1$

$u(-t) = 1$

$u(t) + u(-t) = 2$ but for t=0 only

Gibbs phenomenon : At Point of discontinuity the signal value is given by the average value taken just befor and after the point of discontinuity.

$$\underset{\text{at } t=0}{\text{Value}\left[u(t)+u(-t)\right]} = \frac{\text{after discon} + \text{befor discon}}{2}$$

$$= \frac{1+1}{2} \Rightarrow 1 \quad \text{so} \quad u(t)+u(-t) = 1$$

for $-\infty$ to ∞

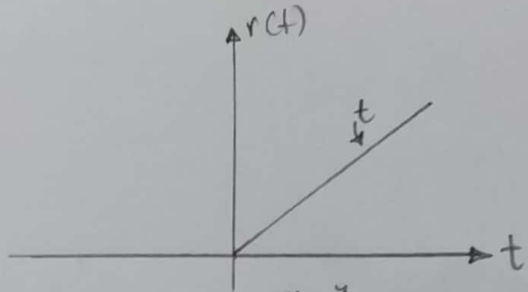
② Unit step signal is Power signal.

③ Unit step signal Neither odd signal No Even signal.

Three: Unit Ramp signal:

$$\rightarrow r(t) = \begin{cases} 0 & t<0 \\ t & t \geq 0 \end{cases}$$

$$r(t) = t \cdot u(t)$$



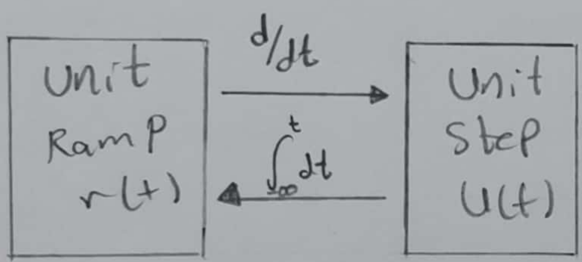
$m = 1$
$c = 0$
$y = mx + c$

Properties of Unit Ramp signal:

① $r(t) = \int_{-\infty}^{t} u(\tau) \cdot d\tau$

because $\int_{-\infty}^{t} u(\tau) d\tau = \cancel{\int_{-\infty}^{0} 0 \, d\tau}^{\text{zero}} + \int_{0}^{t} 1 \, d\tau$

$= [\tau]_0^t$
$= t \cdot 1$
$= t \cdot u(t)$



② Unit Ramp Signal Neither odd Nor Even signal

③ $P = \lim_{T \to \infty} \frac{1}{T} \left[\cancel{\int_{-T/2}^{0} 0 \, dt}^{\text{zero}} + \int_{0}^{T/2} t^2 dt \right]$

$= \lim_{T \to \infty} \frac{1}{T} \left[\frac{t^3}{3} \right]_0^{T/2} \Rightarrow \lim_{T \to \infty} \frac{1}{\cancel{T}} \cdot \frac{\cancel{T^3} T^2}{24} \Rightarrow P = \infty$

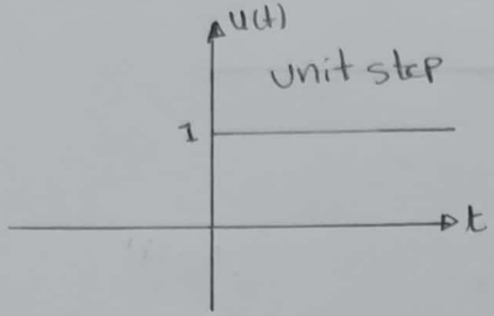
$E = \int_{-\infty}^{\infty} |r(t)|^2 \cdot dt$

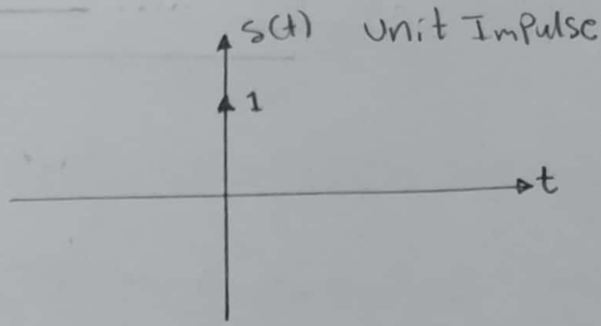
$= \cancel{\int_{-\infty}^{0} 0 \, dt}^{\text{zero}} + \int_{0}^{\infty} t^2 \cdot dt \Rightarrow \frac{1}{3} [t^3]_0^{\infty}$

$E = \infty$

So the Unit Ramp Signal Neither Energy Nor Power signal because $r(t=\infty) = \infty$.

"Relation between Impulse and step signal"



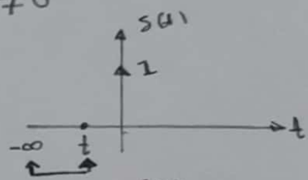


$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$

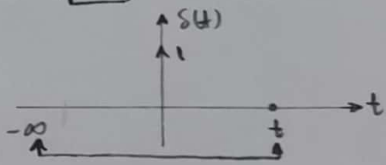
$$\delta(t) = \begin{cases} 1 & t = 0 \\ 0 & t \neq 0 \end{cases}$$

notes:-

$$\int_{-\infty}^{t} \delta(t) \cdot dt = 0 \quad \text{if } t < 0$$



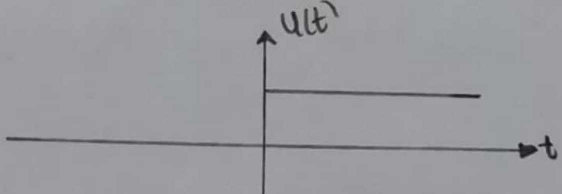
$$\int_{-\infty}^{t} \delta(t) \cdot dt = 1 \quad \text{if } t \geq 0$$
$$= u(t)$$

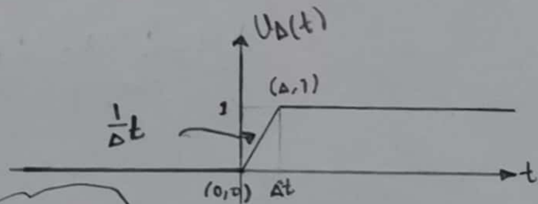


so

$$\boxed{\int_{-\infty}^{t} \delta(t) \cdot dt = u(t)}$$

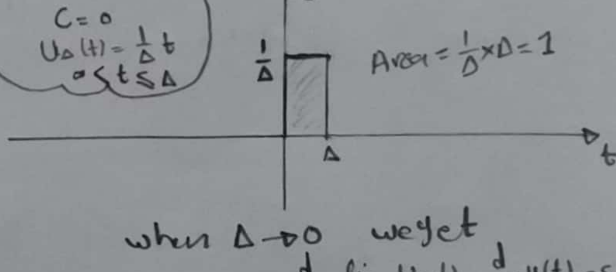
notes:-





$$m = \frac{1}{\Delta}$$
$$y = mx + c$$
$$y = \frac{1}{\Delta}x + c$$
$$c = 0$$
$$U_\Delta(t) = \frac{1}{\Delta}t$$
$$0 \leq t \leq \Delta$$

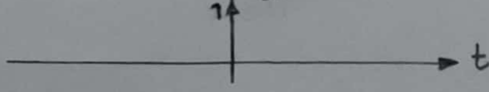
$$\lim_{\Delta \to 0} U_\Delta(t) = u(t)$$



$$\boxed{\frac{d}{dt} \lim_{\Delta \to 0} U_\Delta(t) = \frac{d}{dt} u(t) = \delta(t)}$$

when $\Delta \to 0$ we get

$$\frac{d}{dt} \lim_{\Delta \to 0} U_\Delta(t) = \frac{d}{dt} u(t) = \delta(t)$$



Summary :

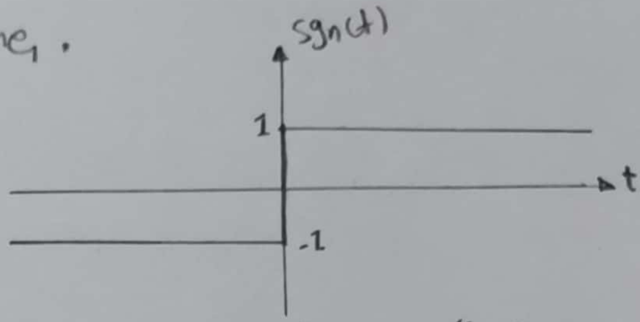
$$\delta(t) \xrightarrow{\int_{-\infty}^{t}} u(t) \xrightarrow{\int_{-\infty}^{t}} r(t)$$

$$\delta(t) \xleftarrow{d/dt} u(t) \xleftarrow{d/dt} r(t)$$

Four : signum Function

→ sign function is another name.

→ $\operatorname{sgn}(t) = \begin{cases} -1 & t<0 \\ 0 & t=0 \\ 1 & t>0 \end{cases}$



at $t=0$ $\operatorname{sgn}(t=0)$ is discontinous.

→ $1+\operatorname{sgn}(t) = 2\,u(t)$

To Prove that

$$\operatorname{sgn}(t) = \begin{cases} -1 & t<0 \\ 1 & t>0 \end{cases} \Rightarrow 1+\operatorname{sgn}(t) = \begin{cases} 0 & t<0 \\ 2 & t>0 \end{cases} = 2\,u(t)$$

so

$$\boxed{1+\operatorname{sgn}(t) = 2\,u(t)}$$

→ $\frac{d}{dt}[\operatorname{sgn}(t)] = 2\frac{d}{dt}u(t)$; $\frac{d}{dt}u(t) = \delta(t)$

i.e $2\delta(t) = \frac{d}{dt}\operatorname{sgn}(t)$

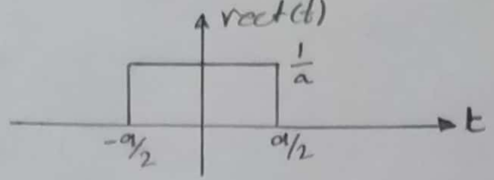
→ $\operatorname{sgn}(t) = \frac{|t|}{t}$ when $t \neq 0$ and $\operatorname{sgn}(t) = 0$ when $t=0$.

→ $\frac{d}{dt}|t| = \operatorname{sgn}(t)$ when $t \neq 0$

Five: Unit Rectangular function:-

→ Another names (Unit Rectangular Pulse) or (Gate function).

→ rect(t) ; OR $\Pi(t)$.



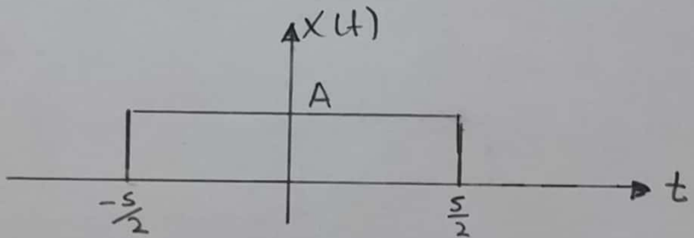
→ Area of Unit Rectangular signal = one because $(\frac{1}{a} \times a) = 1$.

$$\text{rect}(t) = \begin{cases} 0 & t < -a/2 \\ 1/a & -a/2 \le t \le a/2 \\ 0 & t > a/2 \end{cases}$$

→ rect(t) is Even signal and Energy signal.

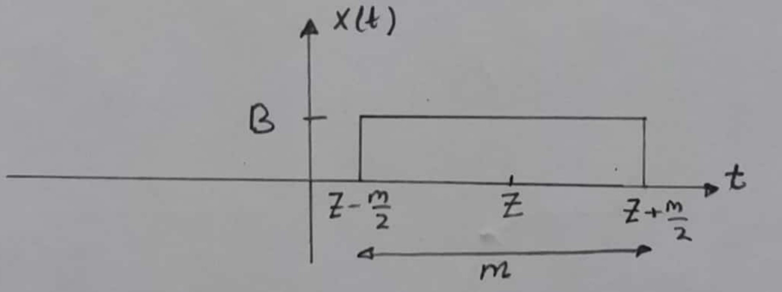
E.X plot the signal $x(t) = A\,\text{rect}\left(\frac{t}{5}\right)$

Sol

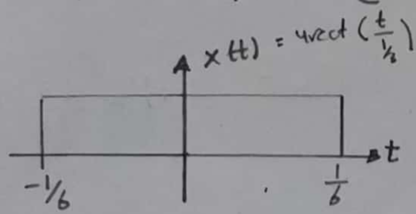


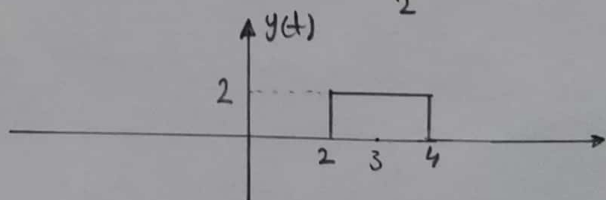
E.X plot the signal $x(t) = B\,\text{rect}\left(\frac{t-z}{m}\right)$ if $z > \frac{m}{2}$

Sol:-



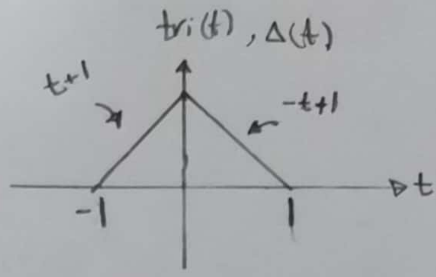
E.X Plot $x(t) = 4\,\text{rect}(3t)$ $\qquad$ $y(t) = 2\,\text{rect}\left(\frac{t-3}{2}\right)$





H.W: Plot $z(t) = \frac{1}{2}\text{rect}\left(\frac{2t-4}{2}\right)$ $\qquad$ $m(t) = \text{rect}\left(\frac{t+\frac{T}{2}}{T}\right)$

Six: Unit Triangular Function:



→ $tri(t) = \begin{cases} t+1 & -1 \le t \le 0 \\ -t+1 & 0 \le t \le 1 \end{cases}$

→ tri(t) or $\Delta(t)$.

→ $\Delta(t) = \begin{cases} 0 & \text{for } |t| \ge 1 \\ 1-|t| & \text{for } |t| < 1 \end{cases}$

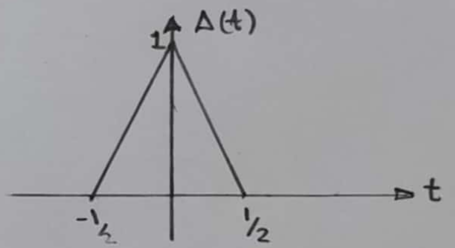
→ Unit Triangular function is Even signal and Energy signal.

⊛ Note that some authors instead define the triangle function to have a base of width 1 instead of width (2):

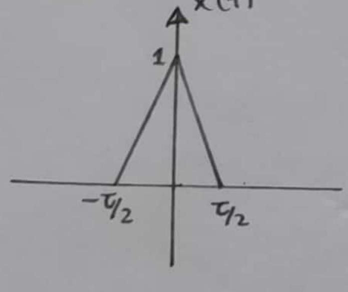
$$tri(t) = \Delta(t) = \begin{cases} 2t+1 & -\frac{1}{2} \le t \le 0 \\ -2t+1 & 0 \le t \le \frac{1}{2} \end{cases} = \begin{cases} 0 & \text{for } |t| \ge \frac{1}{2} \\ 1-2|t| & \text{for } |t| < \frac{1}{2} \end{cases}$$

سوف نعتمد على كون

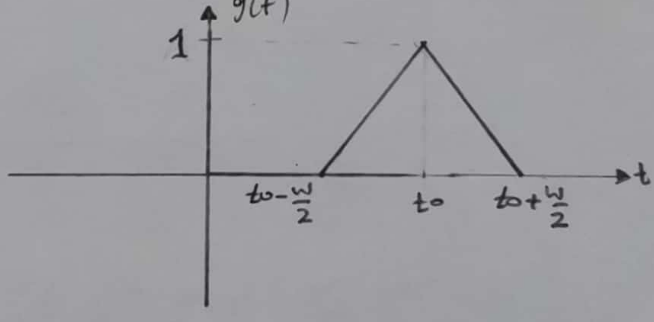
→ $\Delta(t) = tri(t) = \begin{cases} 2t+1; & -\frac{1}{2} \le t \le 0 \\ -2t+1; & 0 \le t \le \frac{1}{2} \end{cases}$



E.X: Plot $x(t) = \Delta\left(\frac{t}{\tau}\right)$



$g(t) = \Delta\left(\frac{t-t_0}{W}\right)$; if $t_0 > \frac{W}{2}$

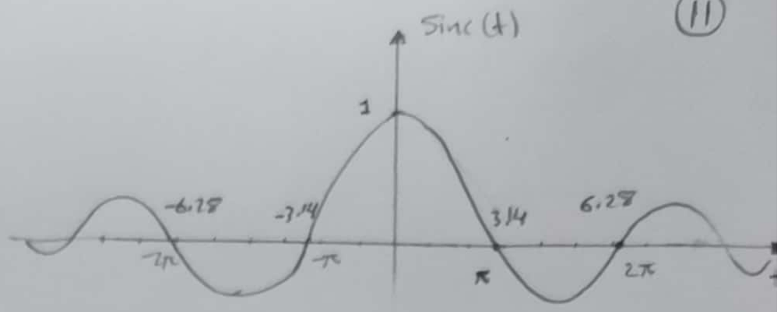


H.W: from above example Find and Plot $y(t) = \frac{d}{dt} g(t)$?

H.W: plot $x(t) = \Delta\left(\frac{t}{2\pi}\right) \cos(10t)$?

Seven: sinc Function:

→ $\text{sinc}(t) = \begin{cases} 1 & t = 0 \\ \dfrac{\sin(t)}{t} & t \neq 0 \end{cases}$



→ $\sin(t) = 0$ only for value of $\sin(t) = 0$ except for $t = 0$

so for $\sin(\mp\pi) = \sin(\mp 2\pi) = \sin(\mp n\pi) = 0$ for $n = 1, 2, 3 \ldots$

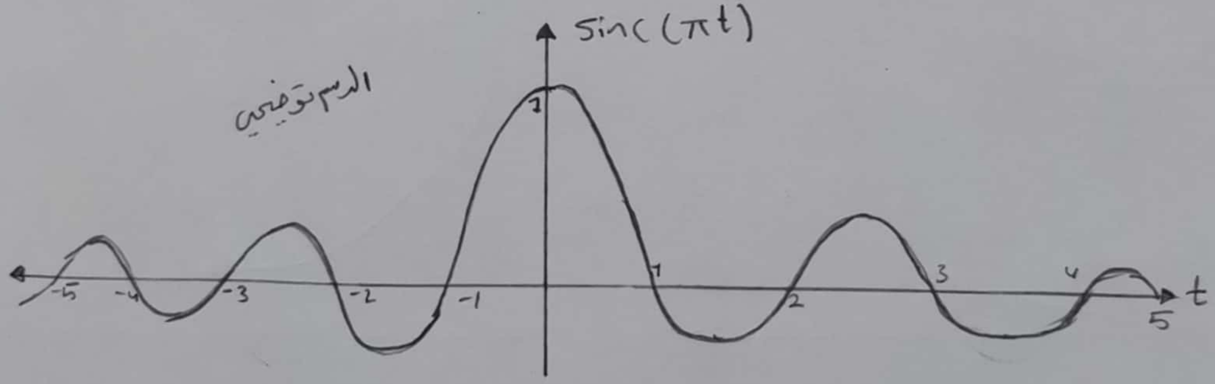
that mean the $\text{sinc}(t) = 0$ for $t = \mp n\pi$ when $\pi = 3.14$

→ sinc function is Even signal and Energy signal.

E.X sketch $\text{sinc}(\pi t)$?

Sol: we know that $\text{sinc}(\pi t) = \dfrac{\sin(\pi t)}{\pi t}$; to get the points at which

$\text{sinc}(\pi t) = 0 \Rightarrow \pi t = \mp n\pi \Rightarrow t = \mp n$ when $n = 1, 2, 3, \ldots$



H.W: Sketch $x(t) = 3\,\text{sinc}(wt)$?

HW: sketch $x(t) = \Delta\left(\frac{t}{2}\right)\text{sinc}(\pi t)$?

Eights Exponential signal

→ $e^0 = 1$ because $e = 2.718\ldots \Rightarrow e^0 = (2.718\ldots)^0 = 1$

→ let $s = \sigma + j\omega$ Complex number then

$$e^{st} = e^{(\sigma+j\omega)t} = e^{\sigma t} \cdot e^{j\omega t} = e^{\sigma t}\left[\cos\omega t + j\sin\omega t\right]$$

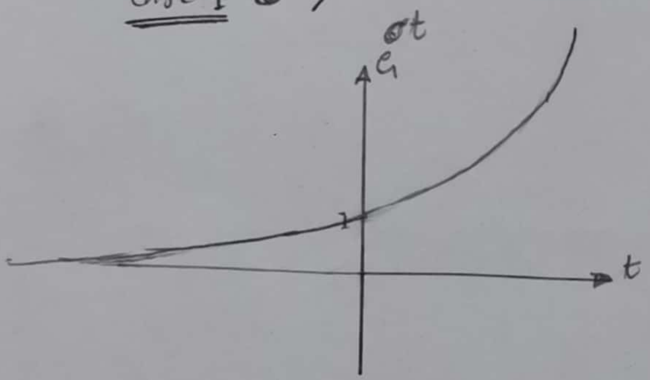
$\boxed{\text{Euler's formula } e^{\mp j\omega t} = \cos(\omega t) \mp j\sin(\omega t)}$

$$e^{st} = e^{(\sigma+j\omega)t} = e^{\sigma t}\left[\cos\omega t + j\sin\omega t\right]$$

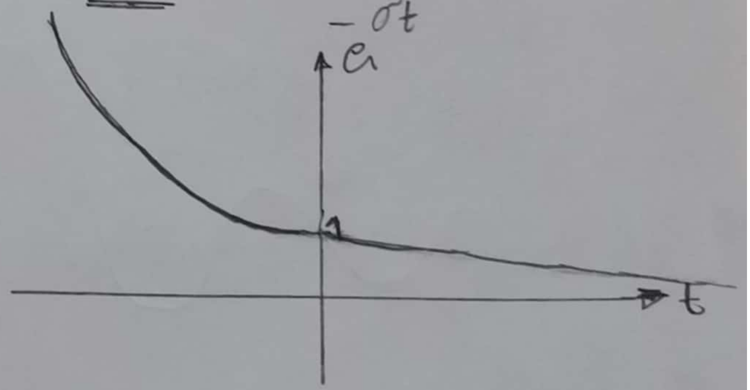
→ for Real Exponential signal $\Rightarrow \boxed{\omega = 0}$ $e^{st} = e^{\sigma t}\left[\cos(0) + j\sin(0)\right]$

$$e^{st} = e^{\sigma t}$$

Case 1 $\sigma > 0$



Case 2 $\sigma < 0$



→ For Complex Exponential signal

Case 1 $\sigma = 0$ & $\omega \neq 0$

$$e^{st} = \cos\omega t + j\sin\omega t$$

Case 2 $\sigma > 0$ & $\omega \neq 0$

$$e^{st} = e^{\sigma t}\cos\omega t + j e^{\sigma t}\sin\omega t$$

Both Real and Imag. Parts will Increasing with time.

Case $\sigma < 0$ & $\omega \neq 0$

$$e^{st} = e^{-\sigma t}\cos\omega t + j e^{-\sigma t}\sin\omega t$$

Both Real and Imag Parts will Decreasing with time

H.W: plot i) $\sin\omega t\, e^{3t}$ ii) $\cos\omega t\, e^{-3t}$

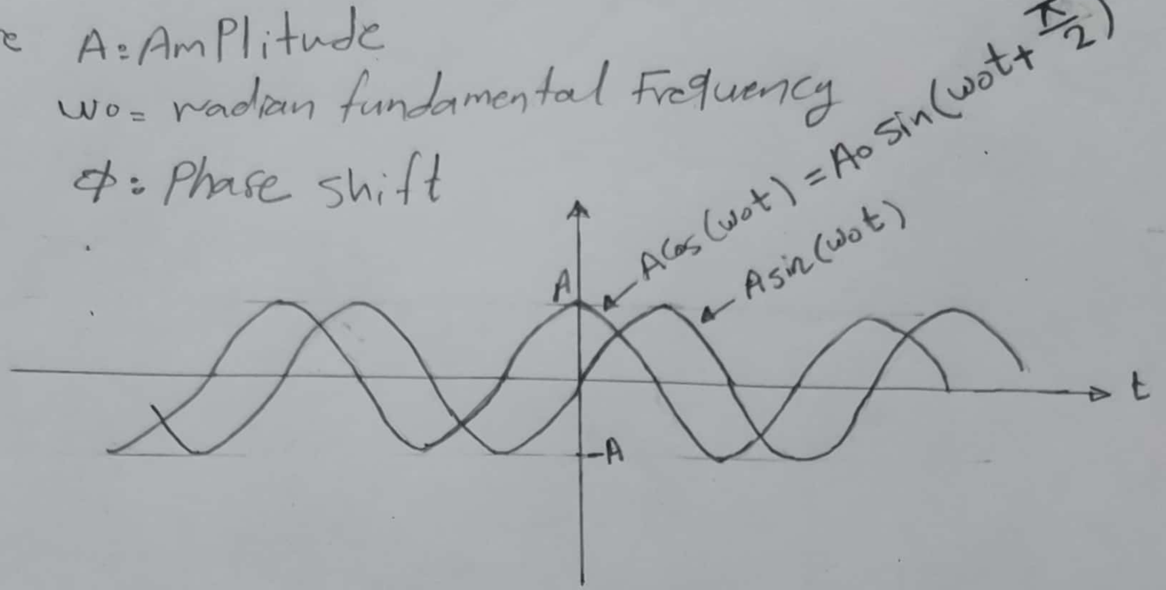
Nine :- Sinusoidal signal :-

→ $A\sin(w_0t \pm \phi)$ OR $A\cos(w_0t \pm \phi)$

where A: Amplitude

w_0 = radian fundamental Frequency

ϕ: Phase shift



→ radian frequency $w_0 = 2\pi f_0$ and $f_0 = \frac{1}{T_0}$ ← Fundamental time Period

→ $\int_{t_1}^{t_2} A\sin(w_0t)\,.dt$ = Zero when $t_1 = -t_2$

→ $\int_{t_1}^{t_2} A\cos(w_0t)\,.dt = 2\int_{0}^{t_2} A\cos(w_0t)\,.dt$ when $t_1 = -t_2$

→ Both $\sin(w_0t)$ and $\cos(w_0t)$ are Power signal

→ $\sin(w_0t)$ is Odd signal, But $\cos(w_0t)$ is Even signal.

"Relation Between Exp and sinusoidal"

① Euler's Formula $e^{j\theta} = \cos\theta + j\sin\theta$

② $\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2} = Re(e^{j\theta})$

③ $\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j} = Im(e^{j\theta})$

H.W: Prove the Euler's Formula ?